MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

COMPARATOR, ANALOG, KEY CODE CFA

This amendment forms a part of Military Specification MIL-M-28787/62C(EC), dated 22 January 1979, and is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

1.2.1, Comparator description, Input offset voltage. Delete "4.0" and substitute "10.0".

1.2.1, Comparator description, Common mode voltage range. Delete "$(+V_{CC}$ -1.3 V)" and substitute "$(+V_{CC}$ -1.5 V)".

1.3.2, delete and substitute:

"1.3.2 Common mode voltage range. The common mode voltage range is between $(-V_{CC}$ +0.4 V) and $(+V_{CC}$ -1.5 V). When the input common mode voltage exceeds the common mode voltage range, the comparator output may latch at a high or low level. Test condition R guarantees correct module operation with a minimum common mode voltage of $(+V_{CC}$ -1.5 V)."

PAGE 3

3.5.8, Marking, second line. Delete "M28787/62" and substitute "M28787/62-1".

PAGE 5

TABLE II, delete test conditions C1-C12, including all columnar data, and substitute:

"	C1-C12	Offset voltage	V	X	-9.0	9.0	-10.0	10.0	-10.0	10.0	-10.0	10.0	mV	"

TABLE II, delete test conditions F1-F12, F13-F18, F19-F24, including all columnar data, and substitute:

"	F1-F6	Output voltage	VIII	X	0.01	0.40	0.01	0.40	0.01	0.40	0.01	0.40	V	
	F7-F12	Output voltage	VIII	X	5.400	5.555	5.400	5.555	5.400	5.555	5.400	5.555	V	
	F13-F18	Output voltage	VIII	S	0.1	1.5	0.1	1.5	0.1	1.5	0.1	1.5	V	
	F19-F24	Output voltage	VIII	S	0.01	0.40	0.01	0.40	0.01	0.40	0.01	0.40	V	"

PAGE 6

TABLE II, delete test conditions R1-R12, including all columnar data, and substitute:

"	R1-R12	Common mode voltage, single supply	XVI	X	2.990	3.020	2.990	3.020	2.990	3.020	2.990	3.020	V	"

PAGES 7 AND 8

4.3, Test equipment, delete in its entirety (including listing) and substitute:

"4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified."

FSC 5963

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:48Z
Check the source to verify that this is the current version before use.

PAGE 9

Add the following new paragraph

"6.3 Cross reference of classification. Part number M28787/62-1 supersedes part number M28787/62. The two parts are interchangeable."

PAGE 15

FIGURE 5: Delete and substitute new figure 5 as printed on page 3 of this amendment.

PAGE 16

TABLE V, Test conditions C1 through C12: Under Min column, delete "-3.0" (two places) and substitute "-9.0" (two places); and under Max column, delete "+3.0" (two places) and substitute "+9.0" (two places).

PAGES 23 AND 24

* FIGURE 8: Delete and substitute new figure 8 as printed on pages 4 and 5 of this amendment.

PAGES 25 AND 26

* FIGURE 9 Delete and substitute new figure 9 as printed on pages 6 and 7 of this amendment.

PAGES 28 AND 29

TABLE VIII: Delete and substitute new table VIII as printed on pages 8 and 9 of this amendment.

PAGE 30

FIGURE 10: Delete and substitute new figure 10 as printed on page 10 of this amendment.

PAGE 31

FIGURE 11, title: Delete and substitute "FIGURE 11. Output voltage test circuit for table VIII (test conditions F13 through F18)."

PAGE 32

FIGURE 12, title: Delete and substitute "FIGURE 12. Output voltage test circuit for table VIII (test conditions F19 through F24)."

PAGE 57

TABLE XVI: Delete and substitute new table XVI as printed on page 11 of this amendment.

PAGE 58

FIGURE 27. Delete and substitute new figure 27 as printed on page 12 of this amendment.

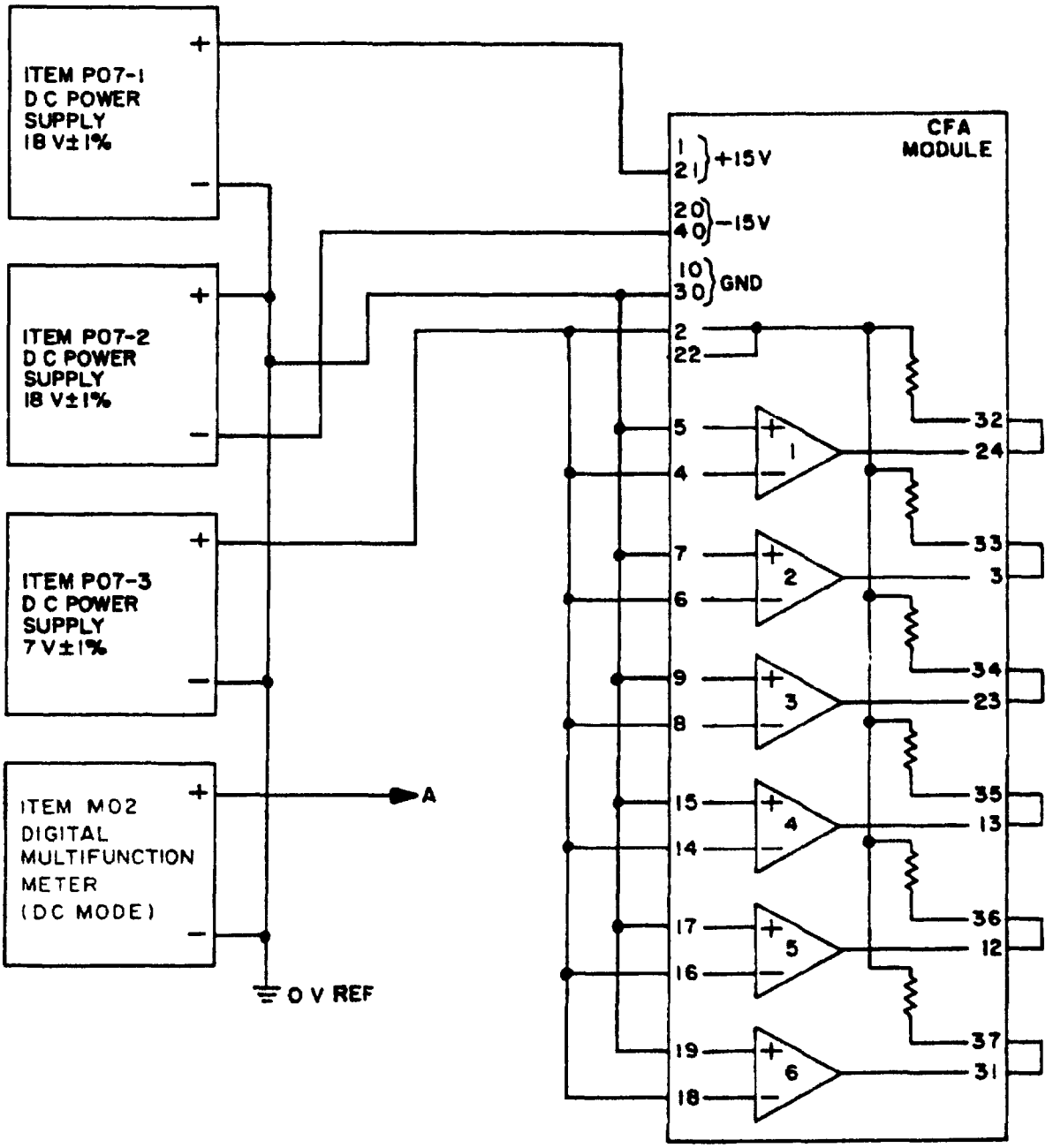
NOTE: The margins of this amendment are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Agent
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N613)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:48Z
Check the source to verify that this is the current version before use.



NOTE: Make the required connections and apply power to the module for 1 minute before making measurements.

FIGURE 5. Stress test circuit for table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:48Z
Check the source to verify that this is the current version before use.

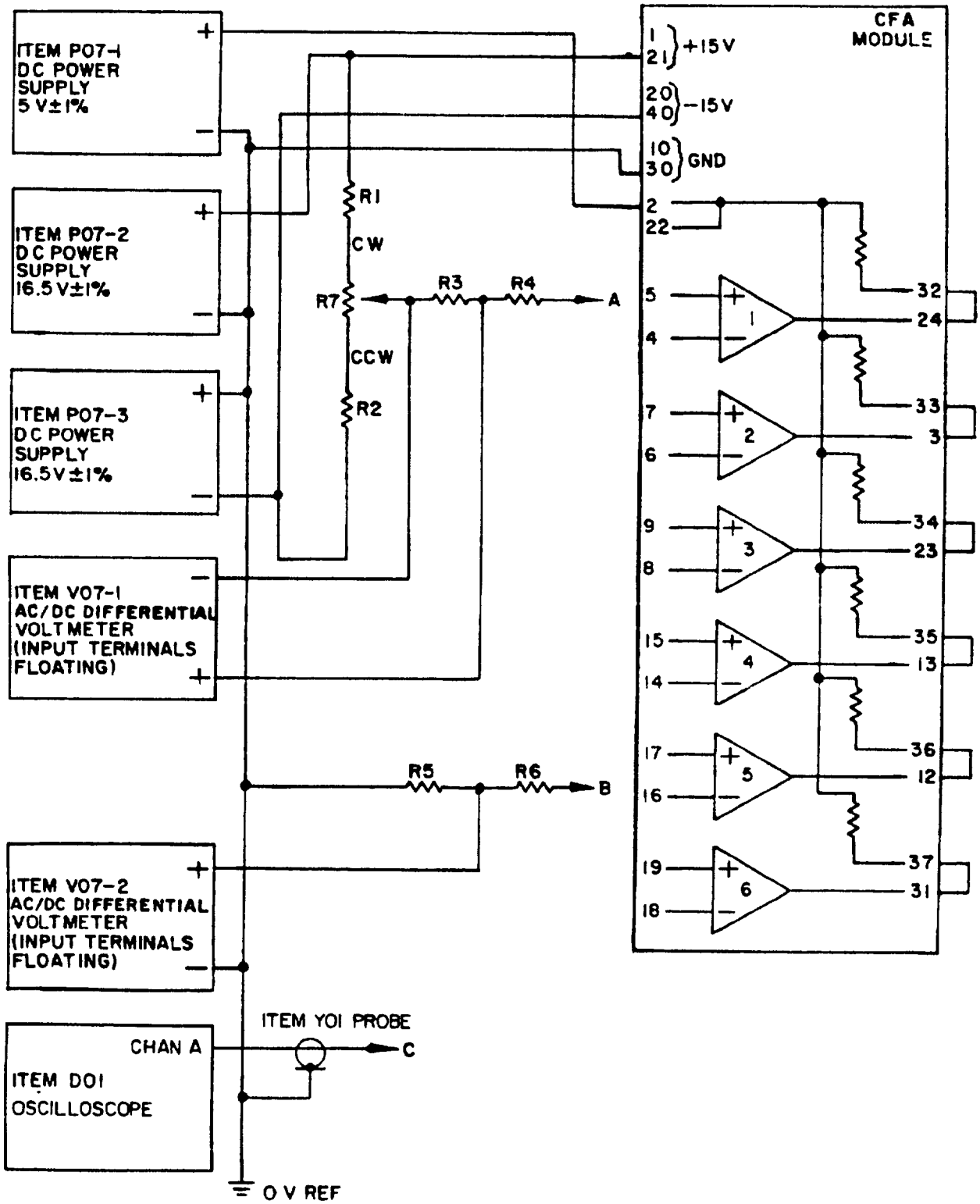


FIGURE 8. Input bias current (±15 V) test circuit for table VI (test conditions D1 through D12).

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:48Z
Check the source to verify that this is the current version before use.

NOTES:
1. R1-R6 = 10.0 kΩ ±1%, 1/8 W.
2. R7 = 500Ω, 0.5 W potentiometer.
3. Before making measurement on each circuit (before test conditions D1, D3, D5, etc.) adjust R7 as follows:
   a. Adjust R7 CCW until the comparator output, as measured with item D01, is between 0 and 1.0 V dc.
   b. Slowly adjust R7 CW until the output switches to a high state (> 4.5 V dc) with no oscillations as verified on item D01.
4. Make a voltage measurement with item V07-1 for test conditions D1 through D11 (odd numbers) and make a voltage measurement with item V07-2 for test conditions D2 through D12 (even numbers). The bias current shall be determined as follows:

$$\text{Bias current} = \frac{V}{10}$$

where V is the voltage in μV
10 is the resistance in kΩ
and Bias current is in nA.

   The resultant shall be within the limits specified.
5. Since the item V07 voltmeters are not used simultaneously only one is required.

FIGURE 8. Input bias current (±15 V) test circuit for table VI (test conditions D1 through D12) - Continued.

Source: https://assist.dla.mil -- Downloaded: 20261O-20T15:48Z
Check the source to verify that this is the current version before use.

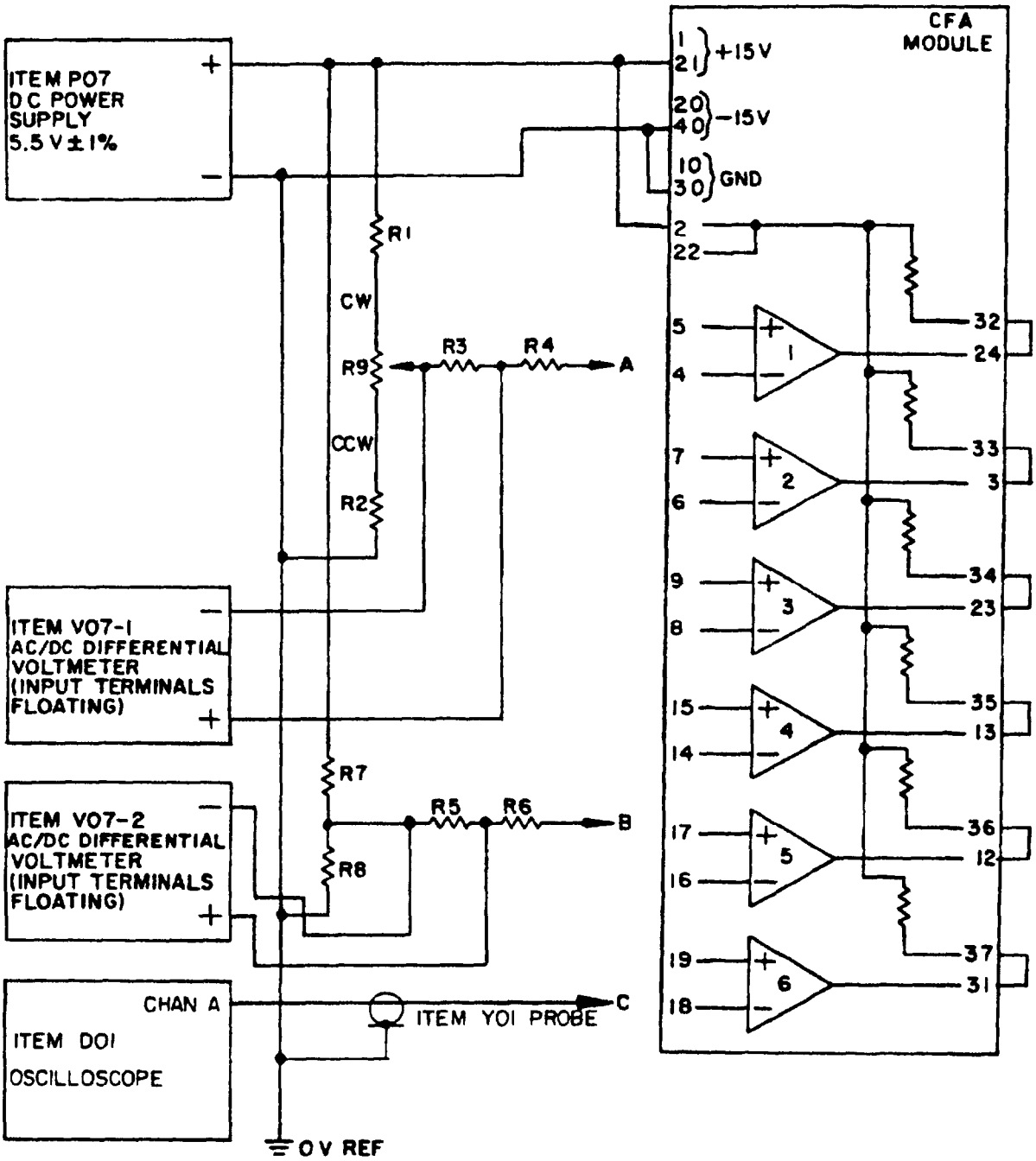


FIGURE 9. Input bias current (+5 V) test circuit for table VI (test conditions D13 through D24).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:48Z
Check the source to verify that this is the current version before use.

NOTES:
1. R1-R8 = 10.0 kΩ ±1%, 1/8 W.
2. R9 = 500Ω, 0.5 W potentiometer.
3. Before making measurements on each circuit (before test conditions D13, D15, D17, etc.) adjust R9 as follows:
   a. Adjust R9 CCW until the comparator output, as measured with item D01, is between 0 and 1.0 V dc.
   b. Slowly adjust R9 CW until the output switches to a high state (> 5.0 V dc) with no oscillations as verified with item D01.
4. Make a voltage measurement with item V07-1 for test conditions D13 through D23 (odd numbers) and make a voltage measurement with item V07-2 for test conditions D14 through D24 (even numbers). The bias current shall be determined as follows:

$$\text{Bias current} = \frac{V}{10}$$

where V is the voltage in μV
10 is the resistance in kΩ
and Bias current is in nA.

   The resultant shall be within the limits specified.
5. Since the item V07 voltmeters are not used simultaneously only one is required.

FIGURE 9. Input bias current (+ 5 V) test circuit for table VI (test conditions D13 through D24) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:48Z
Check the source to verify that this is the current version before use.

TABLE VIII. Output voltage test.

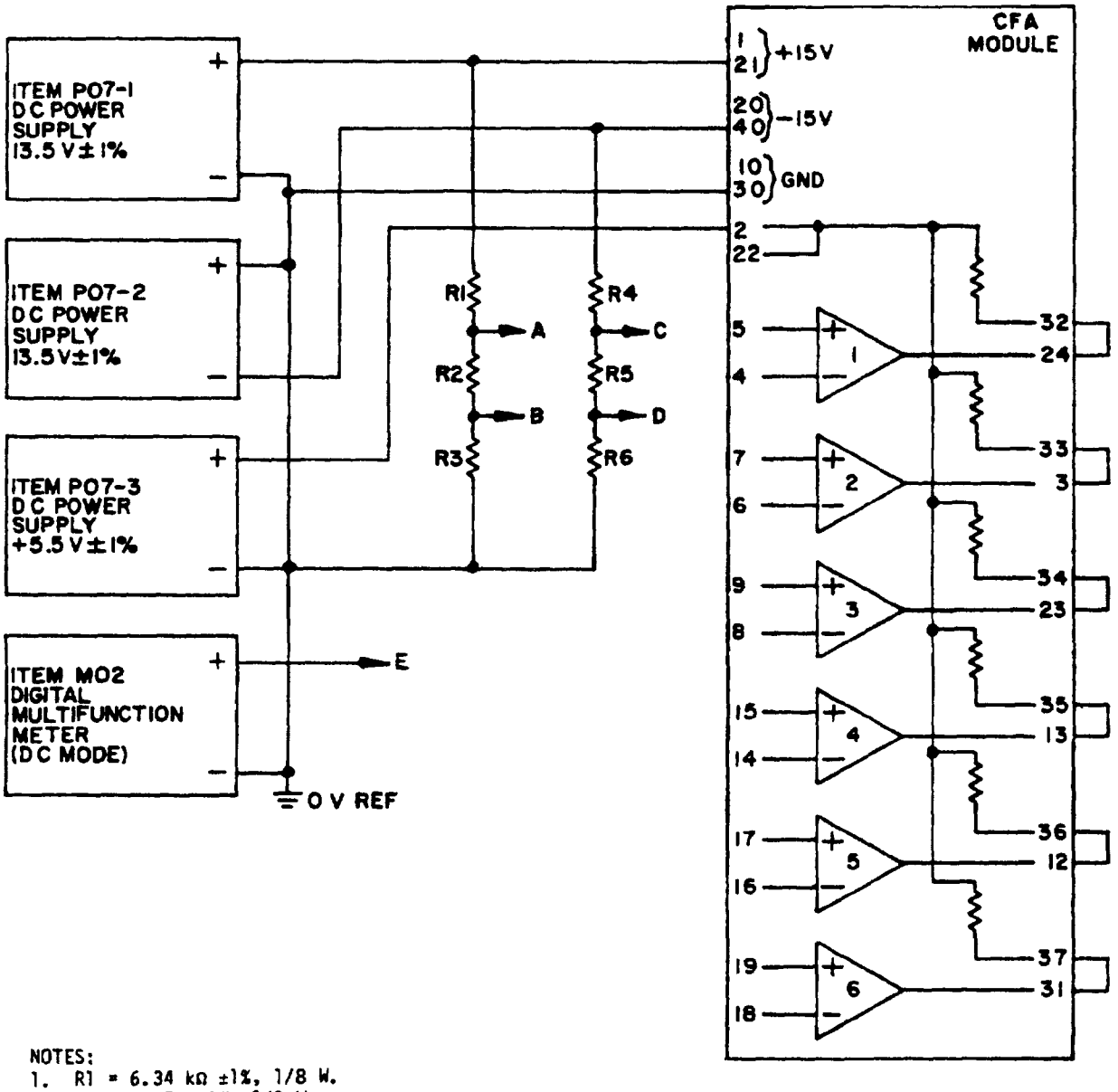
Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
F1	Output voltage I_O = 5.5 mA, V_{IN} = 12.0 Vdc	4	5	Open	Open	24	0.01	0.40	V	10
F2		6	7			3				
F3		8	9			23				
F4		14	15			13				
F5		16	17			12				
F6		18	19			31				
F7		5	4			24	5.400	5.555		
F8		7	6			3				
F9		9	8			23				
F10		15	14			13				
F11		17	16			12				
F12		19	18			31				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:48Z
Check the source to verify that this is the current version before use.

TABLE VIII. Output voltage test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
F13	Output voltage I_O = 50 mA, V_{IN} = 13.1 Vdc	Open	Open	5	4	24	0.1	1.5	V	11
F14				7	6	3				
F15				9	8	23				
F16				15	14	13				
F17				17	16	12				
F18				19	18	31				
F19	Output voltage I_O = 8 mA, V_{IN} = 0.4 Vdc	4	5	Open	Open	24	0.01	0.40		12
F20		6	7			3				
F21		8	9			23				
F22		14	15			13				
F23		16	17			12				
F24		18	19			31				

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:48Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 6.34 kΩ ±1%, 1/8 W.
2. R2 = 26.7Ω ±1%, 1/8 W.
3. R3 = 51.1 kΩ ±1%, 1/8 W.
4. R4 = 1.50 kΩ ±1%, 1/8 W.
5. R5 = 23.7Ω ±1%, 1/8 W.
6. R6 = 49.9 kΩ ±1%, 1/8 W.
7. The voltage at points A and B shall be between 11.86 V and 12.16 V; however the difference between points A and B shall be 6.2 ±0.2 mV.
8. The voltage at points C and D shall be between -13.24 V and -12.97 V; however, the difference between points C and D shall be 6.2 ±0.2 mV.
9. Test conditions F1-F12 verify the correct output voltage and also the common mode voltage range up to ($+V_{CC}$ -1.5 V).

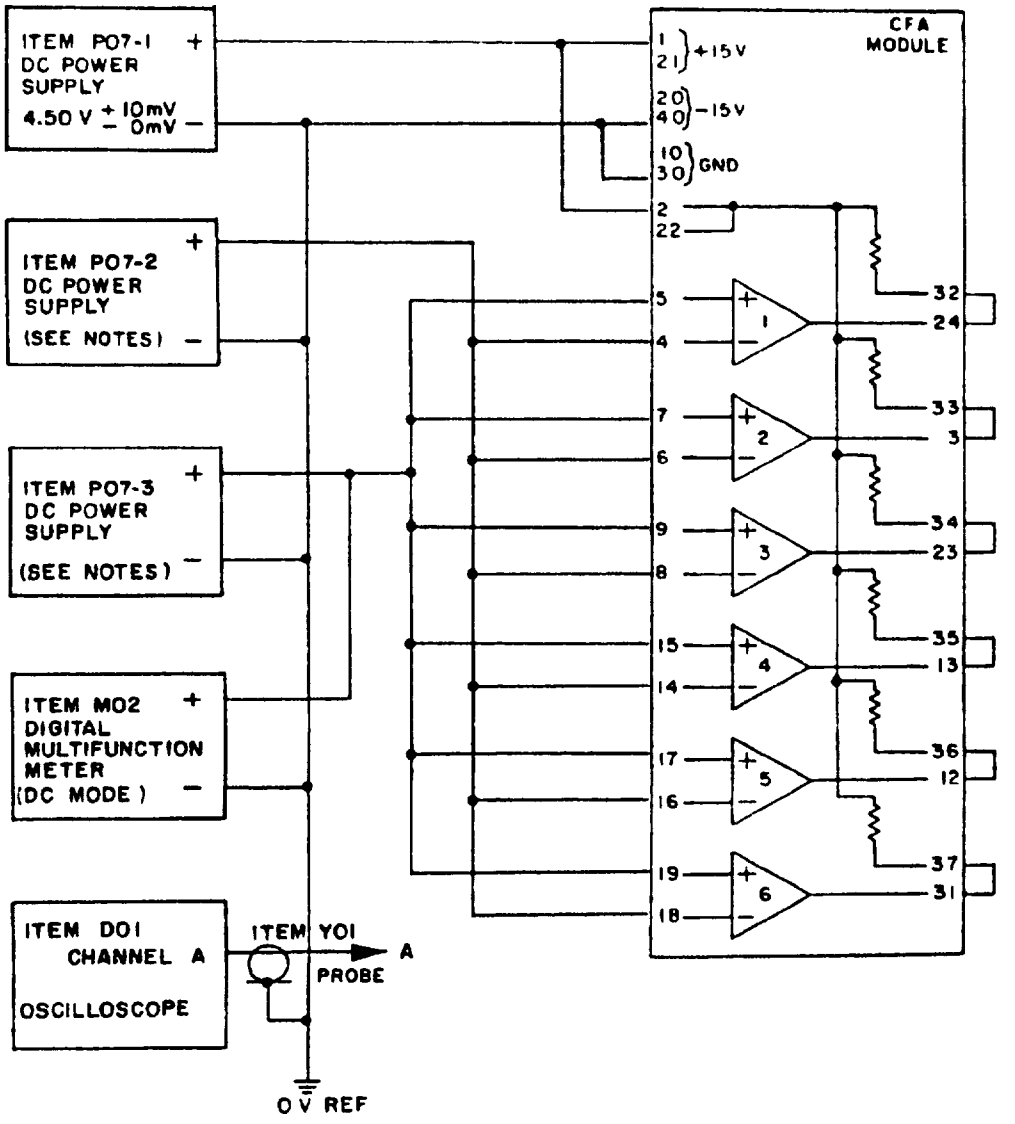
FIGURE 10. Output voltage test circuit for table VIII (test conditions F1 through F12).

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:48Z
Check the source to verify that this is the current version before use.

TABLE XVI. Common mode voltage (single supply) test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
R1	Common mode voltage, single supply	24	2.990	3.020	V	27
R2		3				
R3		23				
R4		13				
R5		12				
R6		31				
R7		24				
R8		3				
R9		23				
R10		13				
R11		12				
R12		31				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:48Z
Check the source to verify that this is the current version before use.



NOTES:
1. For test conditions R1-R6:
   a. Set item P07-2 to 3.00 V +10 -0 mV. Slowly adjust item P07-3 until the output switches to a high state as measured on item D01. The voltage measured on item M02 at this point shall meet the requirements of table XVI.
   b. Before taking measurement, verify that the output high level $\geq$ 4.0 V.
2. For test conditions R7-R12.
   a. Set item P07-3 to 3.00 V +10 -0 mV. Slowly adjust item P07-2 until the output switches to a low state as measured on item D01. The voltage measured on item M02 at this point shall meet the requirements of table XVI.
   b. Before taking measurement, verify that the output low level is $\leq$ 0.8 V.

FIGURE 27. Common mode voltage (single supply) test circuit for table XVI.

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:48Z
Check the source to verify that this is the currentversion before use.